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AI Zealots, Decels, Dancing Monkeys

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4/30/26

What are we doing here

- Dig out entrenched positions
- Validate different perspectives
- Defeat polarization through learning
- Calibrate expectations

What are we NOT doing

- Philosophy: AGI, singularity, or politics.
- Deep technical dive
- Tips / Tricks - rapidly outdated
- Selling you a product

Why listen to me?

- I am **admitting** a long history of being fooled
- I am skeptical, curious, and disagreeable
- I am a huge enthusiast with:
  - little personal responsibilities
  - heavy attraction to opposing viewpoints
  - strong aversion to unfounded doomerism/utopian thinking

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Why not listen to me?

- AI is a wide and rapidly evolving frontier
- I am only one person
- Not an expert in Machine Learning (ML)
- I have a long history of being fooled

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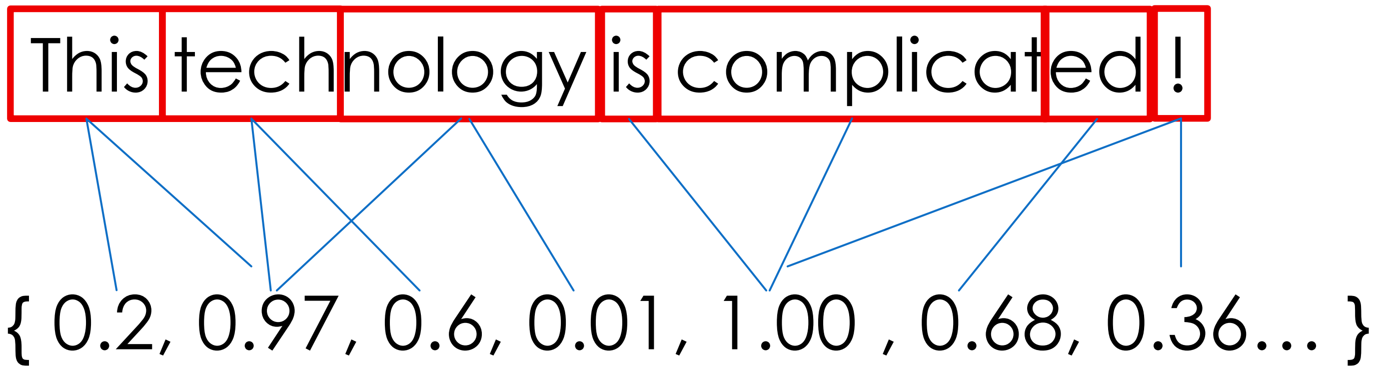
What is an LLM

What LLMs are **NOT**

- Fact database
- Calculator
- Logical analyzer
- Domain Expert

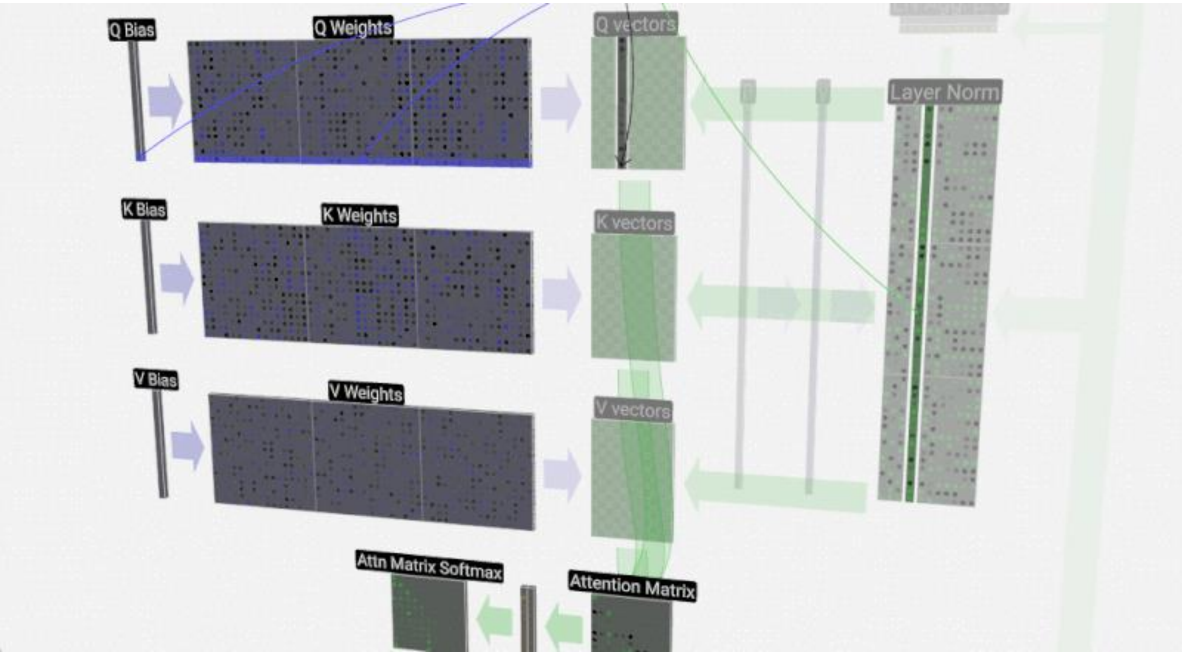
Token

- Chunk of text defined by tokenizer
- Unit of measure for input, processing, output
- Analogous to "lexical" step taken by compilers
- Mapped to vectors of floating point numbers



Large Language Model (LLM)

- Predicts probability distribution over next token
- Neural network based on transformer model
- Composed of layers with billions of floating point numbers (weights)
- Nondeterministic behavior

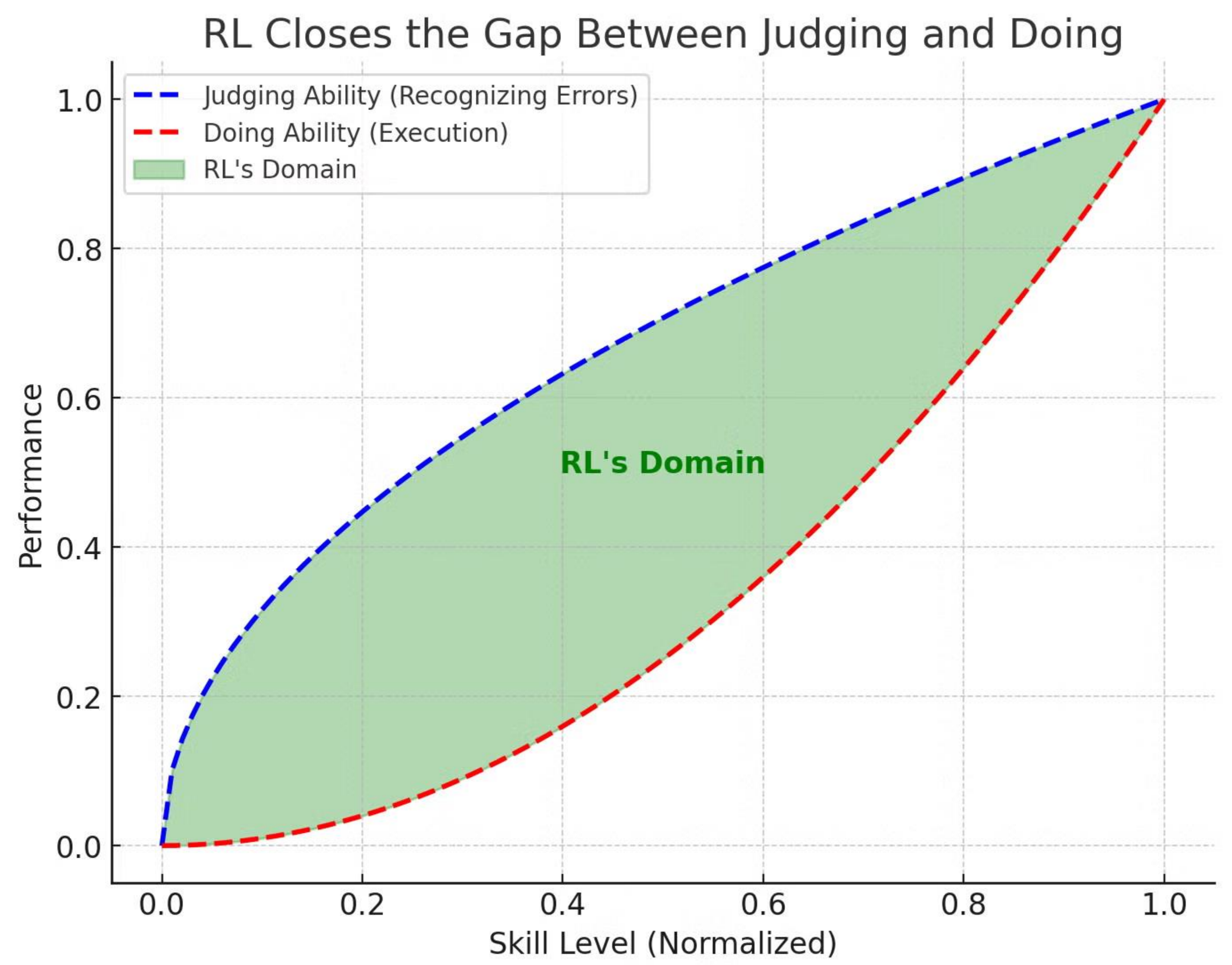


Pre-Training

- Uses input data to refine weights for language and structural patterns
- Back-propagation over trillions of tokens
- Imperfect language word/structure patterns:
  - "how to sound normal"
- Data quality and selection is critical

Post Training

- Fine tuning of model weights
- Optimizing to follow human instructions aligning with human preferences
- Don't answer "what was said", give me "what response was intended"
- Shapes accuracy, tone, and formatting in specific domains
- Methods:
    - Reinforcement learning from human feedback (RLHF)
    - Synthetic data for optimized
    - Safety constraints (e.g. output classifiers)



RL Closes the Gap Between Judging and Doing
Judging Ability (Recognizing Errors)
Doing Ability (Execution)
RL's Domain
RL's Domain
Performance
Skill Level (Normalized)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0

Inference

- Actual use of a model at runtime

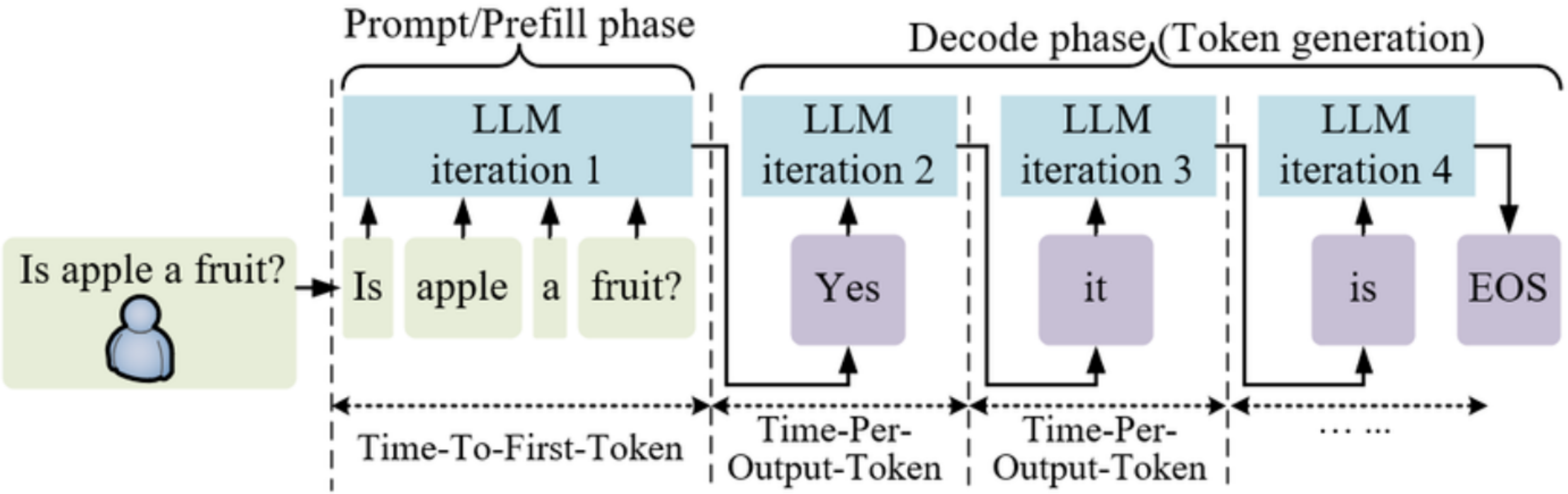
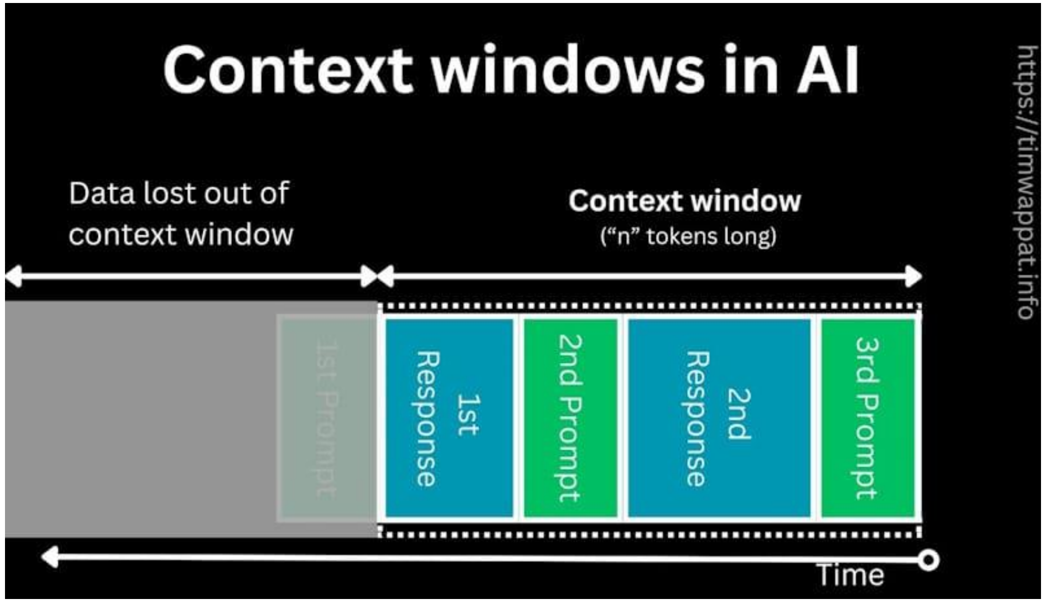


Image Source

Context Window

- Managed "working memory" through attentive sliding token buffer
- The effective "session state" including input + output tokens
- Recent or "identified as practical" tokens are more influential



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Why Now?

Three stars aligned

1. Cloud infrastructure deployed at scale
2. GPU/TPU hardware improvements
3. Breakthrough in neural networks – Transformer architecture

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Who?

Primary US Players

- Claude - Opus, Sonnet, Haiku (Mythos?)
- OpenAI - ChatGPT
- Google - Gemini
- xAI - Grok
- Meta - Llama

Global Players

China

- DeepSeek - DeepSeek v3.2/v4 (MIT license)

France

- Mistral - Mistral mMedium 3 (Apache 2.0 license)

Limitations

Shortcomings

- Non-deterministic behavior
- Weak at math, logic, reasoning
- Domain knowledge is irreplaceable for evaluation
- Limited training data
- Volume of training data affects what appears in models

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Limiting behaviors

- Hallucination:
  - Limited context window.
  - Insufficient confidence assessment.
- Sycophancy:
  - "World traveler" problem.
  - Cost of being wrong and confident.

Ceilings

Data ceiling:

- No more input data to train
- Legal blockers – Copyright
- Generated input - dog-fooding

Material limits

- Realistic token pricing
- Hardware supply chain
- Electricity + water

Algorithm ceiling:

- Hit unsolvable problem

Predictions

Predicting New Technology

- Predictions are often wrong
- Experts disagree
- Reality is often blindsiding

Wrong Prediction Dimensions

- Expansion timeline
- Market segment expansion
- Time to develop, manufacture, deploy
- Harms/Benefits

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The "Obvious Next Step" Throughout History

Flying cars

Prediction:

- Every decade since the 40s: Flying cars are coming in ____ years!

Reality:

- Forever 20-30 years out

AVE Mizar, *Ford Performance and Galpin Motors Archive*

Online Retail

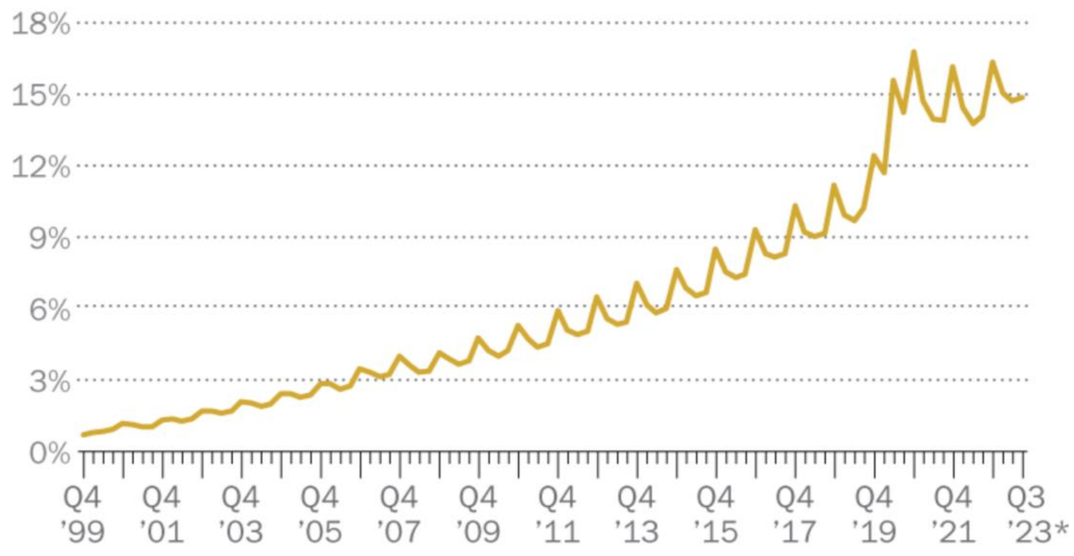
Prediction:

- Nobody will go to the store!

Reality:

- Up from 0.7% in 1999 to 16.7% in 2026

Retail e-commerce sales as a percentage of total retail sales



* Preliminary data as of Nov. 17, 2023.
Note: Figures are not seasonally adjusted.
Source: Quarterly Retail E-Commerce Sales Report, U.S. Census Bureau.

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VR

Prediction:

- Imminent takeover of office work, video games, media

Reality:

- Narrow use for training

What makes you so confident in your predictions, exactly?

- Risk analysis and predictions are difficult
- The field is rapidly changing - many advancements, many opinions
- AI is the most terrible, misleading rollout of any tech product in history.

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Perspective Bias

Perspective Bias

- Everyone has bias
- Nobody is unbiased

Why recognize bias?

- Better calibrate expectations
- Apply appropriate grains of salt
- Temper less warranted fears
- Avoid blindsided harms

Learn

- What to look for
- Who/what to listen to
- How to smell extremes

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Don't believe headlines

- Loudest != informed
- Media selects for shock, anger, hyperbole
- Viral content washes out boring but important developments.
- Telephone effect across layers: Engineer <-> Manager <-> Marketing
- Consider intended audience, speaker bias, technical authority
- Listen to technical experts who use more precision

Case Study: Rust C Compiler

Anthropic – Opus 4.6

- Cost: $20,000 in tokens (2B in, 140M out)
- Can compile:
  - Linux 6.9 – x86, ARM, RISC-V
  - QEMU, FFmpeg, SQlite
  - Doom
- 99% pass on most compiler test suites

Case Study: Rust C Compiler

THE CATCH

- No assembler or linker (uses GCC)
- Does not build ALL programs
- With all optimizations enabled: less efficient than GCC with all optimizations **disabled**

Archetypes

AI Zealot vs Decel

AI Zealot

May be motivated by self interest

- A charlatan (NFT hype)
- Leader seeking investment
- Narcissist
- Gambler

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AI Zealot

May be expert speaking to wrong audience:

- Assumed knowledgable power user
  - skilled at building agents
  - improved prompting techniques
  - describing paid top model performance
- Different inputs -> different outputs
- Often not well communicated

AI Zealot

How are they wrong?

- Directionally correct, but hyperbolic
- Making predictions that are partially right and wrong
- Way in front of the headlights, overly optimistic
- Speaking outside their expertise

Decel

Particular lived experience

- Sees right through hype cycles
- The loud AI pushers are lying so it must all be wrong
- Has made well placed criticisms based on personal experience
- 'it failed so it must be the tool' (not me!)

Decel

Personal preferences

- Strongly favors deterministic systems
- Offended by AI syncophancy
- Generally incurious about best practices, dismissive
- Feels threatened, may not admit it

Decel

How are they wrong?

- The optimal solution is rarely an early iteration
- Relies on long past observations
- "Boy who cried wolf" encourages entrenches positions
- May set arbitrary limits on capability without looking
- Poorly calibrated expectations

Dancing Monkey

- It's the monkey, not the dancing
- Much better dancing exists
- Low bar to impress

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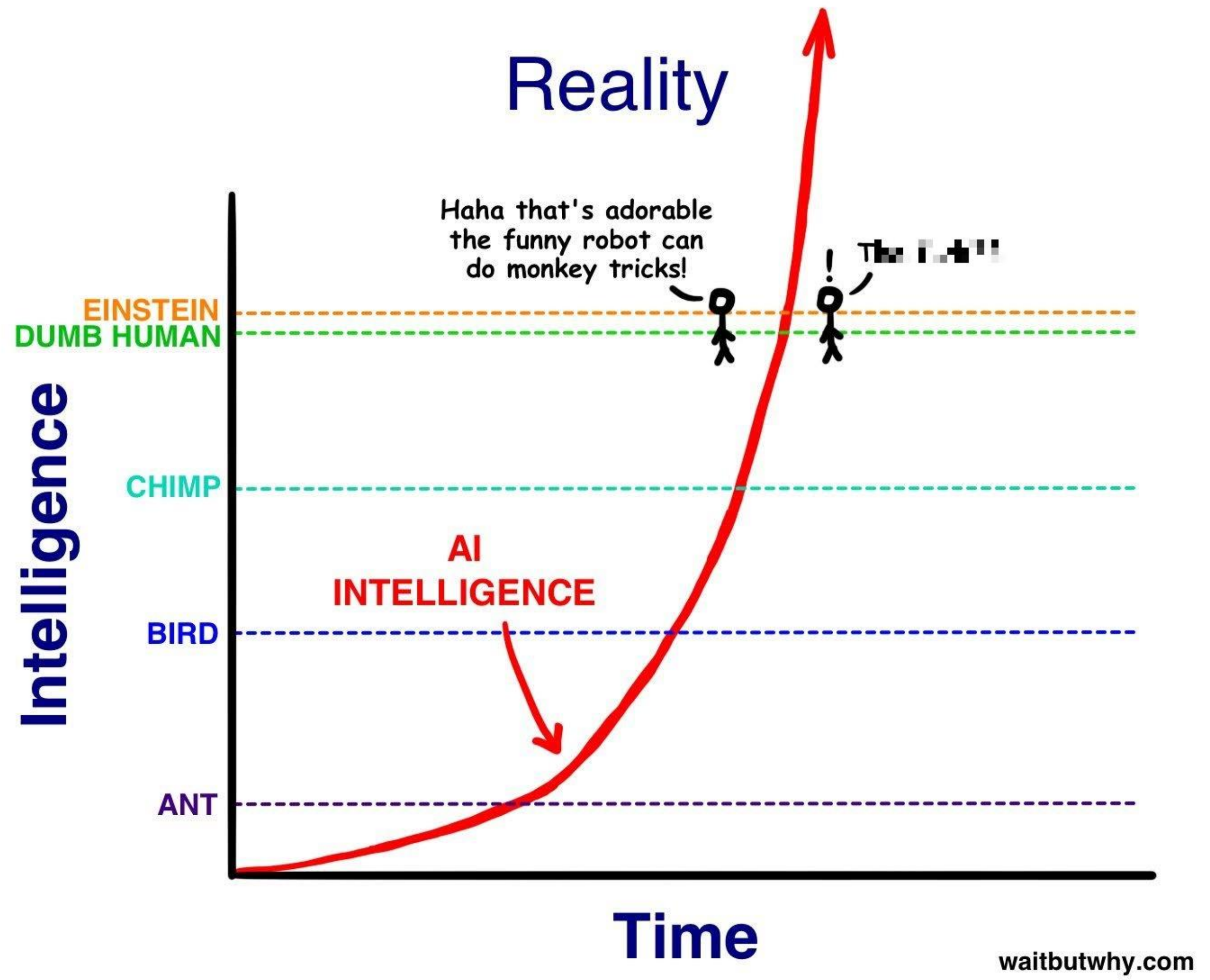
AI Zealot Prediction



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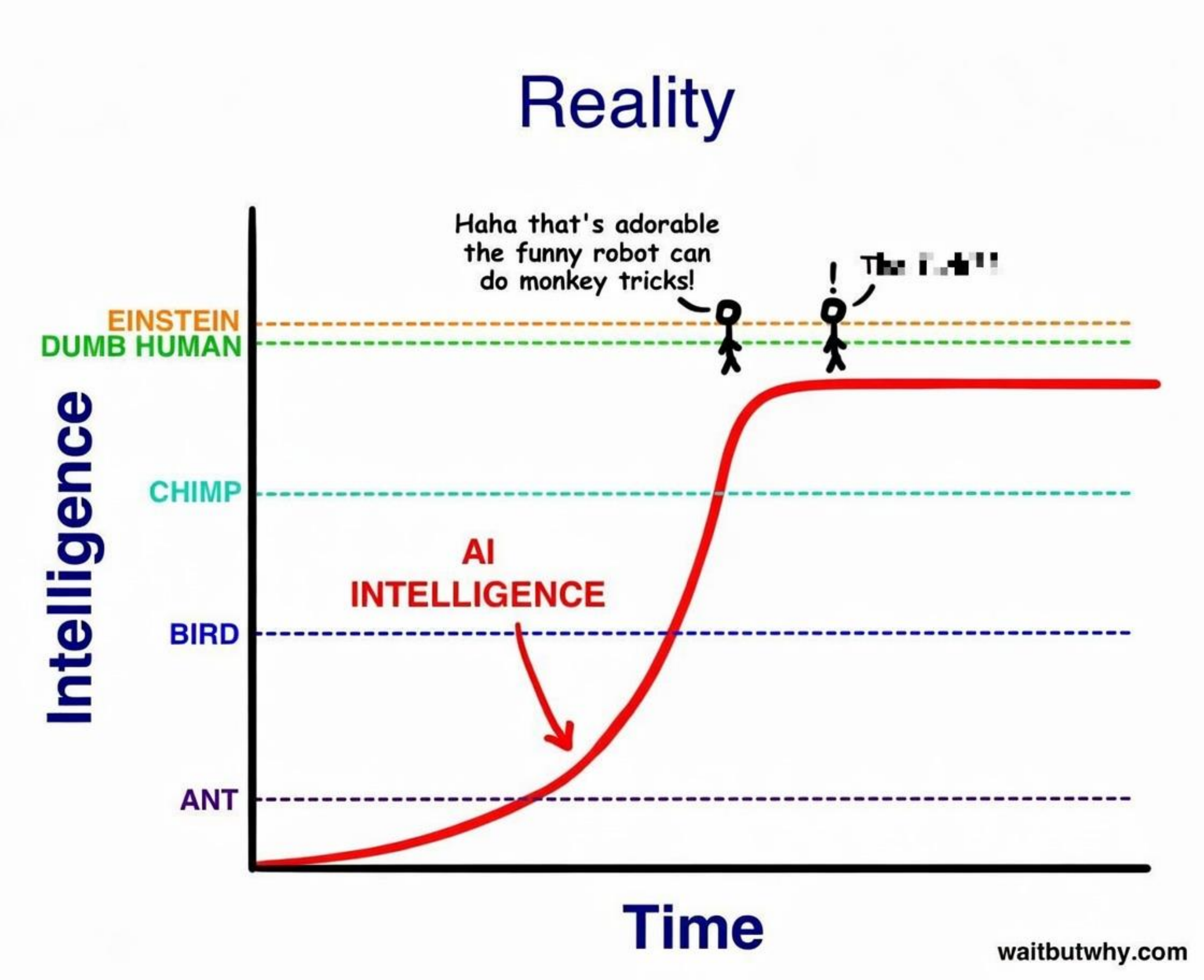
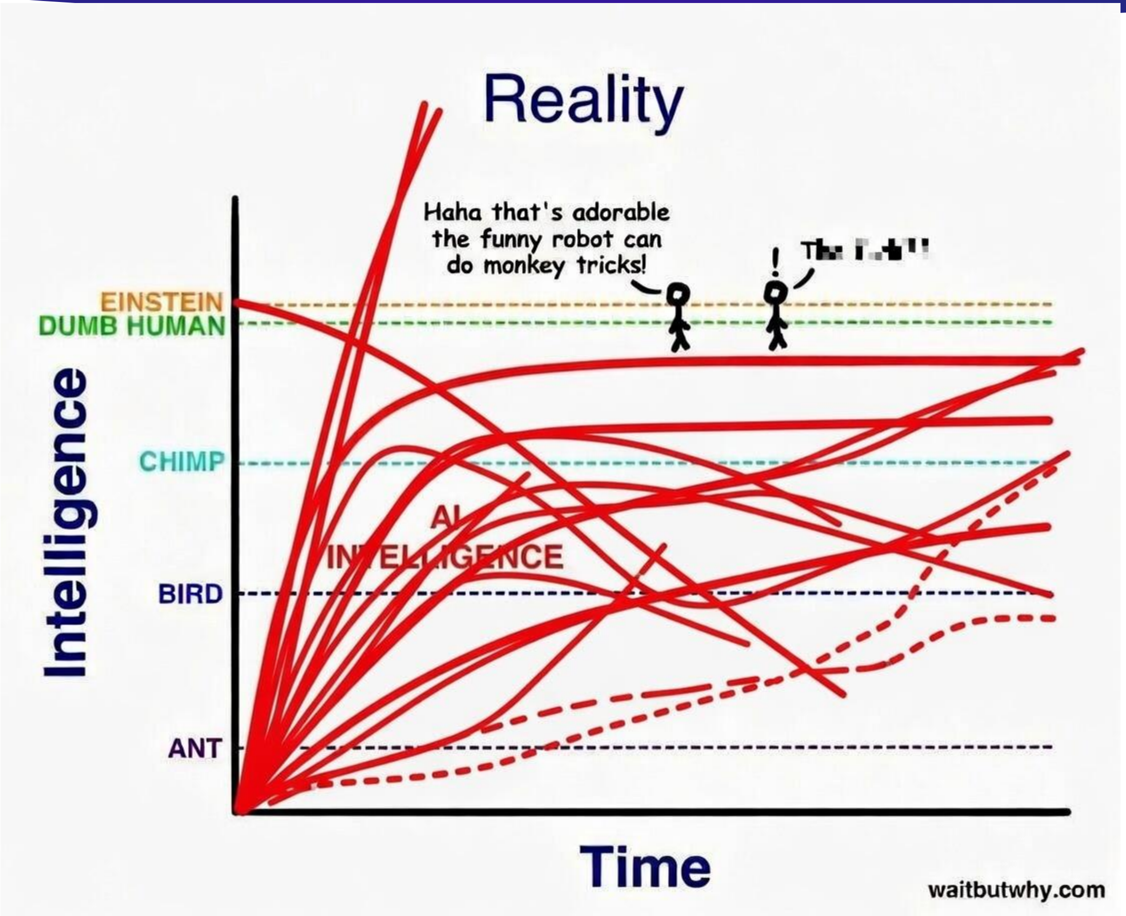
Decel Prediction



My Prediction



What drives bias?

Exposure level matters

- Time makes familiarity and informs expectations
- Direct use improves skill in using a tool more appropriately and effectively
- Enthusiasts use paid models
- Skeptics use free models

Immediate Environment

- Informational asymmetry in exposure to diverse viewpoints
- Varied of quality or volume in each direction.
- Workplace, family, friend circle, media diet

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How to Use More Effectively

Three primary mindsets

1. Vibe coding
2. Augmented learning
3. Genuine enthusiast

- You can slide between, it's allowed
- You don't have to, but knowing helps

1. Vibe coding - What?

- "I just need it to ________"
- Running fast, parallel agents
- Low bar to impress
- May lack expertise or interest to evaluate
- Stop when it works

1. Vibe coding - When?

- Curious but not that curious
- Accuracy or performance not be critical
- Failure has minimal impact
- Interfaces are good, sufficient testing exists

2. Augmented Learning - What?

- Asks for steps, reasoning
- Do not stop when it works
- Stop when you understand enough
- Assume junior engineer capability

2. Augmented Learning – When?

- Limited base knowledge
- Learning curve is very steep
- Dense source material, limited time
- Answering a very specific question
- Search engines suck

3. AI Enthusiast - What?

- Enjoys the act of asking - "Try it again!"
- Asks how the prompt could have been better.
- Iterate on the plan before taking action.
- Constantly trying new tools, techniques
- In it for the love of the game

3. AI Enthusiast - When?

- Want better outputs over time
- Don't want to get left behind
- Time and patience to waste on emerging technology
- It can feel addictive; writing by hand may feel painful, but preserves capability.

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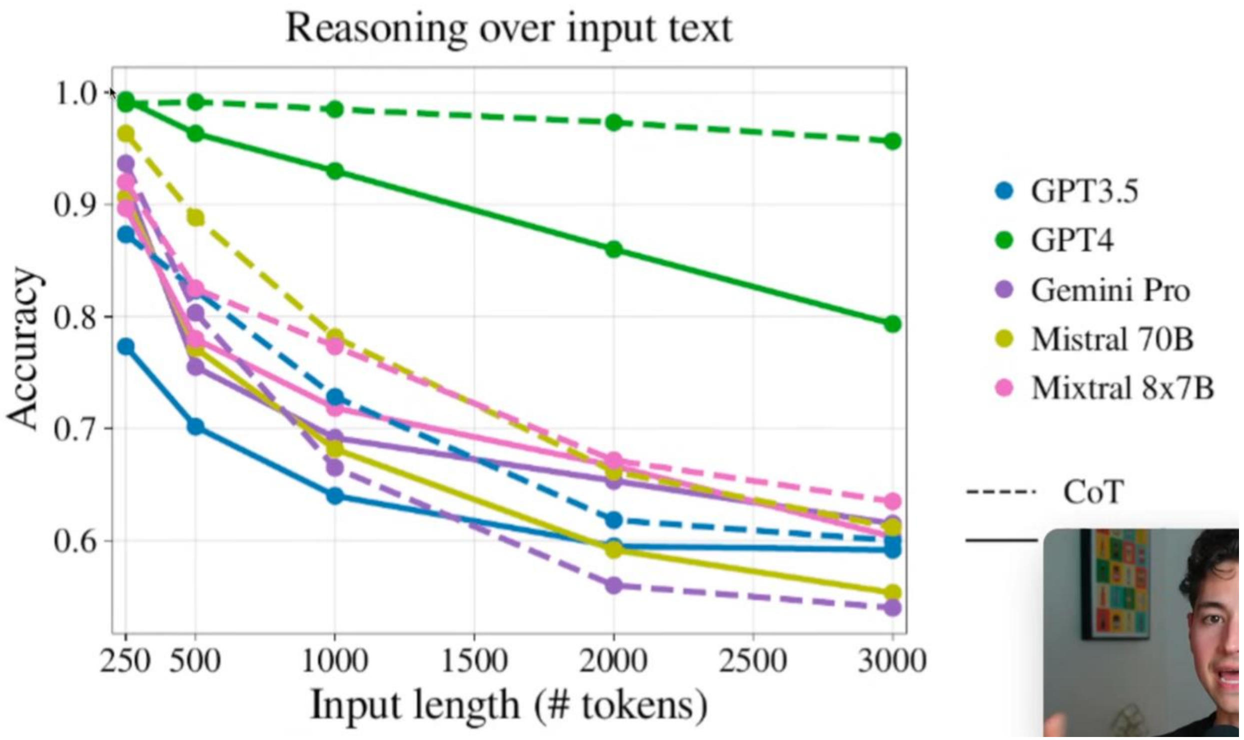
Prompt Engineering

Prompt Engineering

- Use fewer tokens for higher information density.
- Find the Goldilocks zone:
  - Brief and direct
  - Relevant constraints and examples



Prompt Engineering

Write prompts like high-quality Jira tickets:

- Never assume
- Unambiguous and specific
- Include examples
- Define explicit success criteria

Prompt Engineering

- Avoid leading tone
- Present balanced comparison
- Suggested tone:
  - Spartan.
  - Caveman.
  - Researcher.
  - Industry terms.
  - Medium polite.
  - Third person.

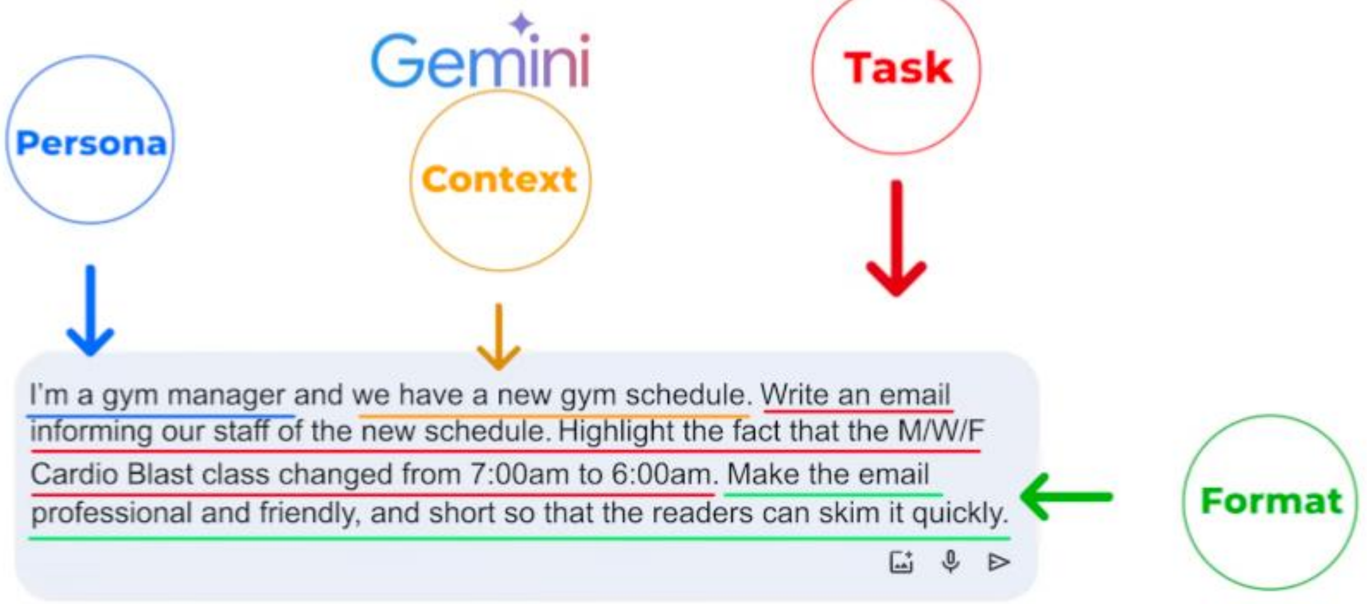
Prompt Engineering

- Give an out when unknown:
  - Ask for sources.
  - Ask for confidence level.
- Avoid "dead zone" prompt drift:
  - Plan A derails
  - Midstream mind changes pollute context
- Avoid contradiction: “Detailed summary”

Prompt Engineering

- Prompt ordering:
  - Put the question first.
- Try different modalities:
  - Add picture or song to enrich narrative.
- Use paid models while exploring:
  - Value your time.
  - Do not bazooka a fly.

Prompt Structure Example



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Risks

Risks

- Cybersecurity:
  - Faster zero-day discovery.
  - Faster cat-and-mouse cycle.
- Bug bounty and application spam overload.
- Inappropriate use cases:
  - Psychology (human isolation risk).
  - Defense, finance, and manufacturing (cannot fail).
  - Fabricated legal cases.
  - PII exposure.

Risks

- Jailbreaking:
  - Universal vs targeted.
  - Constitutional classifiers add compute cost.
- May reflect gender/race bias from training data.
- Reviews can over-prioritize specific concerns; stay aware.

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Long-Term Impacts

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Long-Term Impacts

- Software cost plummets.
  - Graph themes: "more code," "cost per SLOC down," "quality per line down," "KPIs up."
- In less critical applications:
  - Already some slop but "good enough" (daily SQL automation, low-traffic sites).
  - More empowered refactors; stretch goals feel closer.
- Critical industries adopt more slowly.

Long-Term Impacts

- Writing code by hand becomes like guitar:
  - Skill remains.
  - More hobby/craft energy.
- More time in high-level thinking than syntax.
- Accelerated learning/training:
  - Less negative learning.
  - Data is more accessible with context.

Long-Term Impacts

- Lower moat effects:
  - Bring domain knowledge closer to implementation.
  - Break silos.
  - Encourage curiosity.
  - Experience can become a disability; conventional wisdom can become a barrier.
- Strong use cases:
  - Improved medical imaging.
  - Doctors spend more time with people than data entry.

Common Fallacies

STOP USING Anthropomorphic Language

Why?

- Misleading not cute
- Anthromorphic descriptions distort expectation
- Poorly matched expectations are at best annoying

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STOP USING Anthropomorphic Language

BAD	GOOD
he / she / they	it
Thinking	Iterative processing
having a conversation	Generating a transcript role playing a predicted viewpoint

Experts disagree

- This requires a lot of time, effort, and critical thinking to deeply understand.
- Evaluating consequences and contextualization of a problem is highly subjective

Statistics are questioned

- How did who collect what data?
- How much data and where and when?
- Was there selection bias or p-hacking?
- Has this been replicated?

Whitepaper abstract summaries

- What can actually be concluded by the data?
- What exactly is the proportional impact of all of these interconnected causal or correlated factors?
- Any missing controls?
- Conclusions may exclude stronger explanations

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Benchmarks:

- Inexact.
- Hackable.
- Still directionally useful across sets.

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BACKUP SLIDES